THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

CIVL2008 / CIVL2108: PRINCIPLES OF CIVIL ENGINEERING MANAGEMENT

December 14, 2013 14:30 pm – 17:30 pm (3 hours)

Answer **ANY FOUR** questions

All questions carry equal marks

Use of Calculators

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

1. a) As envisioned by the American Society of Civil Engineers, *"... civil engineers serve competently, collaboratively, and ethically as master planners, designers, constructors, and operators of society's economic and social engine ..."*. In the light of this statement, discuss the roles of civil engineers at different project stages to safeguard the best interests of the client and society. [17 marks]

   b) Discuss whether or not it is ethical for the contractors to "form ring" and/or submit a "cover price" and how should consultant civil engineers deal with these situations competently? [8 marks]

2. a) Explain with the aid of suitable diagram(s) the mechanism of "design and build" and "management contracting". [17 marks]

   b) While procurement method is a means to distribute the risks, highlight the risks undertaken by the parties involved in each of the two procurement methods in (a) above. [8 marks]

3. The price breakdown found the lowest tender of a lump sum fixed price contract is shown as follows:

	HK$
Preliminaries	9,500,000
Substructure	11,300,000
Superstructure	62,800,000
External Works	2,700,000
Prime Cost and Provisional Sums	8,100,000
Total:	94,400,000

   a) The contractor has made a qualification in the "Substructure" section of this tender. In order to comply with the original requirement, an additional cost of HK$2 million to the "Substructure" is estimated by the contractor. Explain what is a qualification and how should the engineer respond to the contractor's request. [4 marks]

   b) The engineer has also discovered an arithmetical error in the "External Works" section, and the correct amount for this section should be HK$2.5 million instead. Considering the said error and the scenario in (a), calculate the percentage adjustment. [8 marks]

   c) Explain the purpose of the percentage adjustment computed in (b) and discuss whether or not to upward or downward adjust the tender sum by this percentage. [3 marks]

   d) Describe the two options for dealing with tender errors under the National Joint Consultative Committee's Code of Procedures for Single Stage Selective Tendering. [10 marks]

4. a) Calculate the internal rate of return of an off-shore wind farm based on the following parameters:

Construction cost = $10 billion
Operation period = 20 years
Electricity generated = $2 billion per annum in the first ten years and subsequently at HK$3 billion per annum
Maintenance and operating costs = $0.8 billion per annum
Replacing wind turbine components = $1.5 billion every 6 years
Reversion value = $2 billion [18 marks]

b) Outline the major considerations when the government is evaluating a design-build-finance-operate proposal. [7 marks]

5. a) Explain why it is essential to consider the logical sequence of activities when preparing a network diagram and describe the logic of the four commonly used activity relationships (e.g. finish-to-start relationship) with the aid of suitable example(s). [7 marks]

b) Draw a histogram to show the resource requirement during the project duration based on the data shown in the following table: [15 marks]

Activity	*Start dates*	*Duration*	*Amount in BQ*
A	1 Jul 2014	2.5 months	2 million
B	1 Jul 2014	6 months	7.2 million
C	1 Jul 2014	6 months	3.6 million
D	1 Jul 2014	6 months	2.4 million
E	1 Jul 2014	6 months	4.8 million
F	15 Sept 2014	3.5 months	4.2 million
G	1 Jan 2015	1.5 months	3 million
H	1 Jan 2015	2.5 months	6 million
I	1 Jan 2015	3 months	1.2 million
J	1 Jan 2015	3 months	4.2 million
K	1 Apr 2015	2.5 months	6.5 million
L	15 Jun 2015	0.5 months	0.8 million

c) Comment on the resource profile established in (b) above and suggest measures to eliminate any problem related to resource usage. [3 marks]

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